Linear Algebra II HW 2

Due Date: on or before 18:00, Feb/27/2024 (Thursday), in tutorial.

**1.** Let $\mathsf{T}: \mathsf{R}^2 \to \mathsf{R}^3$ be defined by $\mathsf{T}(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$. Let $\beta$ be the standard ordered basis for $\mathsf{R}^2$ and $\gamma = \{(1,1,0), (0,1,1), (2,2,3)\}$. Compute $[\mathsf{T}]_\beta^\gamma$. If $\alpha = \{(1,2), (2,3)\}$, compute $[\mathsf{T}]_\alpha^\gamma$.

**2.** Define

$$\mathsf{T}: \mathsf{M}_{2\times 2}(R) \to \mathsf{P}_2(R) \quad \text{by} \quad \mathsf{T}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a+b) + (2d)x + bx^2.$$

Let

$$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\} \quad \text{and} \quad \gamma = \{1, x, x^2\}.$$

Compute $[\mathsf{T}]_\beta^\gamma$.

**3.** Let

$$\alpha = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\},$$

$$\beta = \{1, x, x^2\},$$

and

$$\gamma = \{1\}.$$

(a) Define $\mathsf{T}\colon \mathsf{M}_{2\times 2}(R) \to \mathsf{M}_{2\times 2}(R)$ by $\mathsf{T}(A) = A^t$. Compute $[\mathsf{T}]_\alpha$.
(b) Define

$$\mathsf{T}\colon \mathsf{P}_3(R) \to \mathsf{M}_{2\times 2}(R) \quad \text{by} \quad \mathsf{T}(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix},$$

where $'$ denotes differentiation. Compute $[\mathsf{T}]_\beta^\alpha$.
(c) Define $\mathsf{T}\colon \mathsf{M}_{2\times 2}(R) \to R$ by $\mathsf{T}(A) = \operatorname{tr}(A)$. Compute $[\mathsf{T}]_\alpha^\gamma$.
(d) Define $\mathsf{T}\colon \mathsf{P}_3(R) \to R$ by $\mathsf{T}(f(x)) = f(2)$. Compute $[\mathsf{T}]_\beta^\gamma$.
(e) If

$$A = \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix},$$

compute $[A]_\alpha$.
(f) If $f(x) = 3 - 6x + x^2$, compute $[f(x)]_\beta$.
(g) For $a \in F$, compute $[a]_\gamma$.

**4.** For each of the following parts, let $\mathsf{T}$ be the linear transformation defined in the corresponding part of **Q3. Use Theorem 4.2.10** to compute the following vectors:

(a) $[\mathsf{T}(A)]_\alpha$, where $A = \begin{pmatrix} 1 & 4 \\ -1 & 6 \end{pmatrix}$.
(b) $[\mathsf{T}(f(x))]_\alpha$, where $f(x) = 4 - 6x + 3x^2$.
(c) $[\mathsf{T}(A)]_\gamma$, where $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$.
(d) $[\mathsf{T}(f(x))]_\gamma$, where $f(x) = 6 - x + 2x^2$.

**5.** Let $g(x) = 3 + x$. Let $\mathsf{T}: \mathsf{P}_3(R) \to \mathsf{P}_3(R)$ and $\mathsf{U}: \mathsf{P}_3(R) \to \mathsf{R}^3$ be the linear transformations respectively defined by

$$\mathsf{T}(f(x)) = f'(x)g(x) + 2f(x) \quad \text{and} \quad U(a + bx + cx^2) = (a + b, c, a - b).$$

Let $\beta$ and $\gamma$ be the standard ordered bases of $\mathsf{P}_3(R)$ and $\mathsf{R}^3$, respectively.

**(a)** Compute $[\mathsf{U}]_\beta^\gamma$, $[\mathsf{T}]_\beta$, and $[\mathsf{UT}]_\beta^\gamma$ directly. Then verify $[\mathsf{UT}]_\beta^\gamma = [\mathsf{U}]_\beta^\gamma [\mathsf{T}]_\beta$

**(b)** Let $h(x) = 3 - 2x + x^2$. Compute $[h(x)]_\beta$ and $[\mathsf{U}(h(x))]_\gamma$. Then use $[\mathsf{U}]_\beta^\gamma$ from (a) to verify $[\mathsf{U}(h(x))]_\gamma = [\mathsf{U}]_\beta^\gamma [h(x)]_\beta$

6. Let $V$ be a finite dimensional vector space and $T: V \to V$ be a linear transformation. Show that

$$\ker(T) + \mathrm{Im}(T) = V \quad \Leftrightarrow \quad \ker(T) \cap \mathrm{Im}(T) = \{0_V\}.$$

7. Let $A \in M_{n\times n}(\square)$ such that $\mathrm{rank}(A) \le 1$.

(a) Show that there exist $n$-by-1 column vectors $X$ and $Y$ such that $A = XY^T$ and $\mathrm{Tr}(A) = Y^T X$.

(b) Show that $A^2 = \mathrm{Tr}(A)A$.

(c) Suppose $n = 3$. Show that

$$\text{"}A^2 = 0_3 \ (3\times 3 \text{ zero matrix})\text{"} \quad \Leftrightarrow \quad \text{"}\mathrm{rank}(A) \le 1 \quad \text{and} \quad \mathrm{Tr}(A) = 0\text{"}$$